

Stimuli (paired by structure)

NEUTRAL
metalinguistic

FUNC-PFX
programming-style

canonical
op c = `and`

Consider the word
`and` in this sentence.

The function `and(p, q)`
evaluates to true.

invented
op w = `bliq`

Consider the word
`bliq` in this sentence.

The function `bliq(p, q)`
evaluates to true.

Frozen base LM

Gemma 2 9B / OLMo 2 7B / Pythia 6.9B-d

LM forward pass

extract residual stream
at (anchor, layer)

e.g. operator-after, L 4

Linear-probe readout (M-battery)

Fact 1 / M2-canonical

train: NEUT canonical • test: FUNC canonical

= 1.000 [1.000, 1.000]

Fact 2 / M4b

train: NEUT canonical • test: FUNC invented

≠ 0.50 (chance; canonical-set expansion)

Methodological battery (§3.4)

M1

within-condition
5-fold CV

M2-canonical

cross-notation
1-of-K identity

M2-arity

cross-notation
binary-vs-unary

M4b

invented intended-arity
agreement

M4c / pwmin

lucky-default
detector